

Advanced (Habanero)

- Q1.** What is the first step to graph $2x + 3y \geq 12$?
- (A) Graph $2x + 3y = 12$ as a dotted line
 (B) Graph $2x + 3y = 12$ as a solid line
 (C) Determine the y -intercept
 (D) Determine the x -intercept
 (E) Shade above the line
- Q2.** To graph the system $x + 2y \geq 4$ and $3x - 2y \leq 5$, what must be done?
- (A) Graph each inequality on a separate coordinate plane
 (B) Graph the first inequality, then find the intersection with the second inequality
 (C) Graph both inequalities on the same coordinate plane
 (D) Find the intersection point of the two lines
 (E) Shade below both lines
- Q3.** The solution region for the system $y \geq x + 2$ and $y \leq -x + 1$ is
- (A) The area below $y = x + 2$ and above $y = -x + 1$
 (B) The area above $y = x + 2$ and below $y = -x + 1$
 (C) The area above both lines
 (D) The area below both lines
 (E) A single point
- Q4.** The system of linear inequalities $2x + y \geq 3$ and $x - 2y \leq -3$ has
- (A) No solution
 (B) Infinitely many solutions
 (C) A unique solution
 (D) Two solutions
- (E) The solutions are $x = 0$ and $y = 0$
- Q5.** What is the feasible region for $x \geq 0$ and $y \geq 0$?
- (A) The first quadrant
 (B) The second quadrant
 (C) The third quadrant
 (D) The fourth quadrant
 (E) The entire coordinate plane
- Q6.** The inequality $y > x^2 - 4$ has a
- (A) Parabolic boundary
 (B) Linear boundary
 (C) Circular boundary
 (D) Elliptical boundary
 (E) Hyperbolic boundary
- Q7.** To find the vertices of the feasible region for a system of linear inequalities, one must
- (A) Find the x - and y -intercepts
 (B) Find the intersection points of the boundary lines
 (C) Graph each inequality on a separate coordinate plane
 (D) Determine the slope and y -intercept of each line
 (E) Shade above and below each line
- Q8.** The graph of the system $y \geq 2x - 3$ and $y \leq -2x + 1$
- (A) Has a triangular feasible region
 (B) Has a rectangular feasible region
 (C) Is a parabola
 (D) Is an ellipse
 (E) Has no feasible region

Open Ended Questions (FRQ)

- Q9.** Graph the system $x + y \geq 2$ and $x - y \leq 1$. Identify the vertices of the feasible region and shade the region.
- Q10.** The system of linear inequalities is $y \geq x - 2$ and $y \leq -x + 4$. Graph the system, find the vertices of the feasible region, and determine the area of the region.

Chef's Tips

- Tip 1: Emphasize the importance of accurate graphing and shading.
- Tip 2: Ensure students can identify the vertices of the feasible region.
- Tip 3: Provide opportunities for students to practice interpreting the feasible region.